MILITARY SPECIFICATION

RADIATORS, ENGINE COOLING, INDUSTRIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers coolant radiators for liquid-cooled, industrial, internal-combustion engines of 10 horsepower and above.

1.2 Classification. The radiators shall be of the following types, as specified (see 6.2):

Type I - Soldered-tank radiator.
Type II - Bolted-tank radiator with replaceable core.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-L-800 - Lubrication Oil, General Purpose, Preservative (Water-Displacing, Low Temperature).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-C-5501 - Cap and Plug, Protective, Dust and Moisture Seal.
MIL-G-12803 - Gasket Material, Nonmetallic.
MIL-T-22085 - Tape, Pressure-Sensitive, Adhesive, Preservation and Sealing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.
MS35773 - Radiators, Engine Cooling, Industrial.
MS35884 - Cores, Radiators, Engine Cooling, Industrial.

(Copies of specifications, and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM B-36 - Brass Plate, Sheet, Strip, and Rolled Bar.
ANSI/ASTM B-152 - Copper Sheet, Strip, Plate, and Rolled Bar.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A308 - Twine Plates.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The radiators and cores shall be as shown on MS35773, MS35884, and as specified herein.

3.2 First article (first-produced radiator and core). The contractor shall furnish one or more radiators and cores as specified (see 6.2), for examination and tests within the time frame specified (see 6.2), to prove prior to starting production, that his production methods will produce radiators and cores that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests.

3.3 Materials. Materials shall be as specified herein and as shown on applicable standards. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.3.1 Copper tubing. Tubes shall be copper, red brass, or naval brass with 85 percent copper and 15 percent zinc. Tubing shall not be subject to dezincification and resultant corrosion.

3.3.2 Sheet copper. Sheet copper for fins shall conform to ANSI/ASTM B152.

3.3.3 Sheet brass. Sheet brass for tanks and header plates shall conform to ANSI/ASTM B36.

3.3.4 Terne plate. Terne plate for side members for type I radiators shall conform to ASTM A308 Terne Plate, with LT-35 coating thickness as a minimum.

3.3.5 Gasket. Gasket material shall conform to MIL-G-12803, Type I, identification number P1161A.

3.3.6 Dissimilar Metals. Dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact without suitable protection in order to prevent or minimize galvanic corrosion.

3.3.7 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with components and subcomponents. This information shall be available, upon request, to the contacting officer or his or her designated representative.

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3.3.8 Material deterioration and control. The radiators shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.4 Fins and tubes. The core assembly shall be of the tube and plate-fin construction with the fins perpendicular to and in close contact with the tubes.

3.5 Performance.

3.5.1 Heat rejection and core resistance. The heat rejection and core resistance curves of the test core shall be compared with the corresponding heat rejection and core resistance curves of the standard core. A fan curve at standard air density (0.075 lbs/cu. ft.) shall be drawn through the standard core resistance points at the 1500, 1800 and 2100 feet per minute air velocity points. The point at which the fan curve intersects the test core resistance curve shall be projected onto its respective heat rejection curve. When compared to the standard core values, the projected core values shall conform to the values stated on sheet A of MS35773.

3.5.2 Heat rejection comparison. The comparison between heat rejection (air gain) and heat rejection (water loss) shall not exceed 5 percent.

3.5.3 Distortion. When tested as specified in 4.5.2.3, the radiator and core shall show no leakage or permanent distortion exceeding 1/8 inch.

3.5.4 Vibration and leakage. The radiator or core when vibrated as specified in 4.5.2.4 at the most critical resonant frequency shall show no evidence of structural damage, seepage, or leakage.

3.6 Identification marking. Radiators and type II cores shall be identified in accordance with MIL-STD-130. A metal identification plate shall be permanently affixed to the side of the top tank of each radiator near the water inlet. In addition, a metal identification plate shall be permanently attached to the side of the type II core. Marking shall include the following information:

Military standard radiator
MS part number
Manufacturer's name or trademark

3.7 Finishing and painting. Unless otherwise specified (see 6.2), the radiator and core shall be finished and painted in accordance with the contractor's standard practice.

3.8 Type I. The type I soldered-tank radiator shall be as shown on MS35773, part numbers -1, -2, -3, -9 or -10, as specified (see 6.2).

3.9 Type II. Bolted-tank radiator with replaceable core, type II, shall be as shown on MS 35773, part numbers -4, -5, -6, -7, or -8, as specified (see 6.2).

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3.10 Workmanship.

3.10.1 Cleaning. The completed radiator and cores shall be cleaned and free from defects such as dirt, sand, metal chips, rust, cracks, and other foreign materials or defects that could impair their serviceability. Water from the cleaned radiator or cores shall not show a "pH" change of greater than 1, the chloride content shall not exceed 75 parts per million, nor shall there be a change in chlorine content no greater than 75 parts per million when compared with water before rinsing.

3.10.2 Soldering. Soldering of the overflow tube to the filler neck and the filler neck to the radiator top tank shall not interfere with the proper operation of the pressure cap. In the event any soldering is done to parts after the radiator or core has been cleaned, the radiator or core shall be cleaned again after the soldering operation.

3.10.3 Header plates. The gasket area, the corresponding boltholes, and the bolthead area of the header plates used on type II radiators shall be free of solder lumps and a water resistant nonhardening cement shall be used on the gaskets when the core is assembled to the tanks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Failure of any test by the first-produced model shall be cause for disassembly, in the presence of a Government representative, of the first-produced model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First-produced radiator inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First-produced radiator inspection.

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4.3.1 Examination. The first-produced radiator(s) shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection of all first-produced radiator(s).

4.3.2 Tests. The first-produced radiator(s) shall be tested as specified in 4.5.2.1 through 4.5.2.5. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 through 4.5.2.5. AQL shall be 2.5 percent defective for major defects and 4 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The radiators or cores shall be examined for the following defects:

Major

101. Dimensions not as specified.
102. Material not as specified.
103. Soldering not as specified.
104. Materiels are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environments.
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
106. Contractor does not have documentation available for identification of material, material finishes, or treatments.

Minor

201. Identification marking incorrect or illegible.
202. Cleaning not as specified.
203. Workmanship not as specified.

4.5.2 Tests.

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4.5.2.1 Conditions. The following conditions shall apply during the tests performed in accordance with this specification on complete radiators and on radiator cores when they are furnished separately. Radiator cores furnished separately shall be tested in fixtures which simulate, for test purposes, the top and bottom radiator tanks.

4.5.2.1.1 Supports. The radiator or core shall be supported on its normal points of support and shall not be supported on a cradle or bracket which in any way restrains the possible distortion of the radiator or core when under pressure.

4.5.2.1.2 Equipment. The test radiator or core, as applicable, shall be mounted on a test stand. A circulating pump shall be used to maintain an even temperature in the water reservoir. A flow pump and piping shall be used to provide water to the test radiator or core and a throttling valve shall be used to regulate the flow to the desired rate. A variable speed fan or adjustable dampeners or louvers shall regulate the airflow to the test radiator or core.

4.5.2.1.3 Instrumentation. Instrumentation shall be provided to perform the following functions:

a. To measure the temperature in the waterline at inlet and outlet to test radiator or core.
b. To measure airflow in the airduct. (When an orifice is used to measure the airflow, a manometer shall be connected upstream from the orifice to indicate the static pressure in the duct.)
c. To measure the pressure drop or the resistance to airflow across the core.
d. To measure the temperature of the airflow on each side of the core.

4.5.2.1.4 Control limits and data observations. The observed data shall be recorded. All points for each test shall be recorded only after all variables have been stabilized. The degree of stabilization and accuracy of observations shall be acceptable if the heat-rejection comparison conforms to 3.5.2.

4.5.2.1.5 Coolant. The coolant shall be water.

4.5.2.1.5.1 Coolant temperatures. The temperature of the water entering the test section shall be between 170° F and 210° F.

4.5.2.1.6 Heat rejection. The heat rejection test shall be made using one of the following test methods.

4.5.2.1.6.1 Heat rejection, test method (a). The core section shall be tested in the wind tunnel at the 100 percent waterflow rate and at 125 percent of the rated waterflow. If these selected values are not possible the core shall be tested at not less than three waterflow rates which bracket the 100 and 125 percent rate for each radiator size. At each waterflow rate, the heat rejection shall be determined at not less than four air velocities overlapping the range of 1,500 to 2,100 feet per minute as indicated on sheet 7 of MS35773. The heat

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rejected by the water and the heat gained by the air shall be separately calculated at each test condition. A performance curve shall be plotted as shown on MS35773, Figure 1 or 2, as applicable. The heat rejection and core resistance values at 1,500, 1,800, and 2,100 feet per minute air velocity shall be taken from the performance curve and recorded. The difference between the test data and standard core data shall be checked for conformance to 3.5.1.

4.5.2.1.6.2 Heat rejection, alternate test method (b). When the same core section is used for more than one radiator type and size, waterflow rates and air-velocity rates which overlap the entire range of required conditions shall be chosen for test points on the sample core. All the air velocities shall be tested at each chosen waterflow rate. These results shall be plotted as shown on MS35773, figure 1 or 2, as applicable. The varying waterflow rates in gallons per minute can be plotted as a parameter on these curves. From these curves, a cross plot can be made with heat rejection in Btu/minute as the ordinate, waterflow rates as the abscissa, and air velocity lines of 1,500, 1,800, and 2,100 feet per minute as the parameter. From this cross plot, the required values can be obtained and recorded. The data thus obtained shall be compared to the standard core values of MS35773, Figures 1 and 2, as applicable. The difference between the test data and standard core data shall be checked for conformance to 3.5.1.

4.5.2.1.7 Air flow. Air flow through the test core section shall be from either side.

4.5.2.1.8 Air-pressure-drop-corrections. The air-pressure-drop measurements shall be corrected to standard conditions by use of the following formula:

$$\Delta P_c = \frac{(\Delta p)\rho}{\rho_o}$$

ΔP_c = Corrected drop, in H_2O.

Δp = Measured drop, in H_2O.

ρ = Inlet air density, Lbm/cu. ft.

ρ_o = Standard air density, 0.075 Lbm/cu. ft.

4.5.2.1.9 Vibration. The radiator shall be filled with tapwater for the test specified in 4.5.2.4. The radiator shall be supported as specified in 4.5.2.1.1 and securely fastened to a rigid mounting bracket which shall be bolted to the vibration table to insure that the motion of the radiator shall be essentially the same as the motion of the platform. Means shall be provided for controlling the direction of vibration of the test machine and for adjusting and measuring frequencies and amplitudes of vibration to keep them within prescribed limits.

4.5.2.2 Heat rejection and core resistance. The radiator or core, as applicable, shall be tested as specified in 4.5.2.1.6 based on rated heat rejection as defined in 6.4.12. These values may be determined on a square-foot basis. These values shall then be corrected to the area in square feet. Nonconformance to 3.5.1 shall constitute failure of this test.

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4.5.2.3 Pressure cycling. The complete radiator shall be tested with all outlets closed. Pressure, variable from atmospheric to 150 percent of cap pressure not exceeding 18 psig, shall be applied at the inlet using air, steam or glycol and shall be maintained at a temperature of at least 212° F during the test. When steam is used, means shall be provided to prevent the accumulation of water. The pressure cycling shall take place in 3 to 4 seconds and at a rate of 6 pressure cycles plus or minus 1 cycle per minute. The radiator shall be cycled to a minimum of 50,000 pressure cycles. The radiator shall be examined periodically for evidence of leakage or distortion. Only tube leaks and tube-to-header leaks, not to exceed three, shall be repaired or plugged before continuing the test. Any evidence of leakage or maximum permanent distortion of more than 1/8 inch shall constitute failure of this test.

4.5.2.4 Vibration.

4.5.2.4.1 Resonance survey. The radiator which has been tested in accordance with 4.5.2.3 may be rebuilt or another radiator may be used for this test. The radiator shall be tested for leakage prior to start of this test. The radiator shall be prepared in accordance with 4.5.2.1.9 and vibrated at frequencies from 10 cps to 33 cps at the table displacements specified in table I. The change in frequency shall be made at intervals of 1 cps and maintained at each frequency for 1 minute. If resonance occurs at any point in the specified range, the frequency of vibration at that point shall become the test condition for 4.5.2.4.2.

4.5.2.4.2 Procedure. This test shall be run after completion of 4.5.2.4.1 if no leaks are observed. The radiator shall be vibrated for not less than 24 hours at the most critical resonant frequency. If no resonance was observed, this test shall be performed at 33 cps at the displacement referenced in table I. The radiator shall be periodically examined during the test for evidence of seepage and leakage. Nonconformance to 3.5.4 shall constitute failure of this test.

Table I. Table Displacements.

Frequency range	Total table displacement
cps	inches
10 to 15	0.060 ± 0.006
16 to 25	0.050 ± 0.005
26 to 33	0.040 ± 0.005

4.5.2.5 Washing operation. The radiator or core shall be filled with distilled water or tapwater of known pH index (see 6.4.13), at a temperature of 80° F to 100° F. The radiator shall then be shaken to assure thorough mixing and then allowed to stand for 1 hour. At the end of the 1-hour period, 1/2 gallon of the contents shall be drained into a clean container and tested for pH change and chloride content. Nonconformance to 3.10.1 shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

104.	Materials, methods, or containers not as specified for level A or B. Each incorrect material, method or container shall constitute one defect.
105.	Openings into radiators or replacement core assemblies not sealed as specified for level A.
106.	Gaskets not preserved as specified for level A.
107.	Contents not immobilized within box as specified for level A.
108.	Box closure and sealing not as specified for level A.
109.	Radiators or replacement core-assemblies of unlike description packed together for level A or B.
110.	Quantities packed together exceed the weight limitation of the box for level A or B.
111.	Preservation or packing not in accordance with the referenced document as specified for commercial.
112.	Marking missing, illegible, incorrect, or incomplete for level A, B or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2) radiators or replacement core-assemblies shall not require application of a contact preservative. When specified (see 6.2), slush or flush interior of radiators replacement core-assemblies with VV-L-800 to insure complete coverage and thoroughly drain excess. Openings into radiators or replacement core-assemblies shall be sealed with tape conforming to MIL-T-22085, Type II or with caps or plugs of the appropriate size conforming to MIL-C-5501. The filler neck cap shall be secured in place to prevent loss. Gaskets for each replaceable core assembly shall be protected with fiberboard stiffeners and shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3. Each radiator or each replaceable core-assembly with gaskets, shall then be placed in a close-fitting box conforming to PPP-B-636, class weather resistant, grade as applicable, style optional. The contents shall be blocked, braced or cushioned as applicable within the box to prevent movement or damage. Box closure and sealing shall be as specified for method V in the appendix to the box specification.

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5.1.2 Commercial. Each radiator or replacement core-assembly with gaskets, shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, level B or commercial as specified (see 6.2).

5.2.1 Level A. Radiators of like description or replacement core-assemblies of like description, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional, in quantities not to exceed the weight limitation of the box. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Radiators or replacement core-assemblies with gaskets shall be packed as specified in 5.2.1 for level A except that boxes shall be domestic type.

5.2.3 Commercial. Radiators or replacement core-assemblies with gaskets, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The radiators covered by this specifiction are intended for use with liquid-cooled internal combustion engines to dissipate the heat from the coolant.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. Type of radiator or core, when applicable, required (see 1.2).
d. Time frame required for submission of first-produced radiator(s) and number of radiators required (see 3.2).
e. When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. Finishing and painting when other than as specified (see 3.7).
g. Applicable MS part number required (see 3.8).
h. Applicable MS part number required (see 3.9).
i. Degree of preservation and degree of packing required (see 5.1 and 5.2).
j. When interior of radiator or replacement core-assemblies are to be preserved (see 5.1.1).

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6.3 First-produced radiator. Any changes or deviations of production radiators and cores as applicable, from the approved first-produced radiator during production will be subject to the approval of the contracting officer. Approval of the first-produced radiator will not relieve the contractor of his obligation to furnish radiators and cores as applicable, conforming to this specification.

6.4 Definitions. For the purpose of this specification, the following definitions shall apply.

6.4.1 Rated internal pressure. The rated internal pressure shall be 7 psig for radiators conforming to MS35773-1 through -10.

6.4.2 Standard air. Standard air shall be air at a temperature of 70° F, a barometric pressure of 29.92 inches of mercury, and a density of 0.075 pound per cubic foot.

6.4.3 Rated air velocity. The rated air velocities shall be those listed on sheet 7 of MS35773.

6.4.4 Rated coolant flow. The rated coolant flow shall be the flow listed on sheet 7 of MS35773, and shall be based on approximately a 10° F drop in coolant temperature in the radiator.

6.4.5 Heat transfer - heat energy absorbed by air flow. The heat energy absorbed by air flow is the product of the air flow, specific heat of air, and air temperature rise.

Heat energy absorbed by air flow = $WC_p \Delta T$

W = air flow rate, Lbm/Min.

C_p = 0.24 Btu/Lbm/°F

ΔT = air temp rise, °F

6.4.6 Heat transfer - the heat rejection from water flow. The heat rejection from water flow is the product of the coolant flow, specific heat of coolant and coolant temperature drop.

Heat transfer (heat rejection from water flow) = $WC_p \Delta T$

W = water flow rate, Lbm/Min.

C_p = 1 Btu/Lbm/°F

ΔT = water temp drop, °F

6.4.7 Specific heat at constant pressure of water. The specific heat at constant pressure of water shall be based on the average water temperature. The

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specific heat at constant pressure shall have the value of 1 Btu/lb./° F for water temperatures up to 215° F.

6.4.8 Specific heat at constant pressure of air. The specific heat at constant pressure of air shall have the value of 0.24 Btu/lb./° F for air temperatures up to 215° F and pressure around 1 (atm.).

6.4.9 Average water temperature. The average water temperature of the radiator is the sum of the water inlet and outlet temperatures divided by 2.

6.4.10 Rated potential. Rated potential shall be defined as the average water temperature minus the entering air temperature and shall have a value of 80° F.

6.4.11 Observed potential. Observed potential shall be defined as the observed average water temperature minus the observed entering air temperature during heat-rejection and core-resistance test.

6.4.12 Rated heat rejection. The rated heat rejection shall be total observed heat rejection in Btu's per minute multiplied by the rated potential and divided by the observed potential.

6.4.13 Index of acid intensity. The pH index shall be defined as the negative logarithm of the hydrogen ion concentration per liter. A pH index of 7 is a neutral solution. A pH index of less than 7 is an acidic solution. A pH index of greater than 7 is a basic solution.

6.4.14 Resonance. Resonance is a condition of maximum magnification of an applied vibration. It is usually manifested by visibly increased vibration of the radiator under test.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

Custodians:
Army - ME
Air Force - 99

Review activity:
Army - AT
Air Force: 82
DLA - CS

Preparing activity:
Army - ME

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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